HDOC Day-70 – C Programming

1. Aim

To write a C program to calculate the digital root of a number using recursion and use a static variable to count the number of recursive calls.

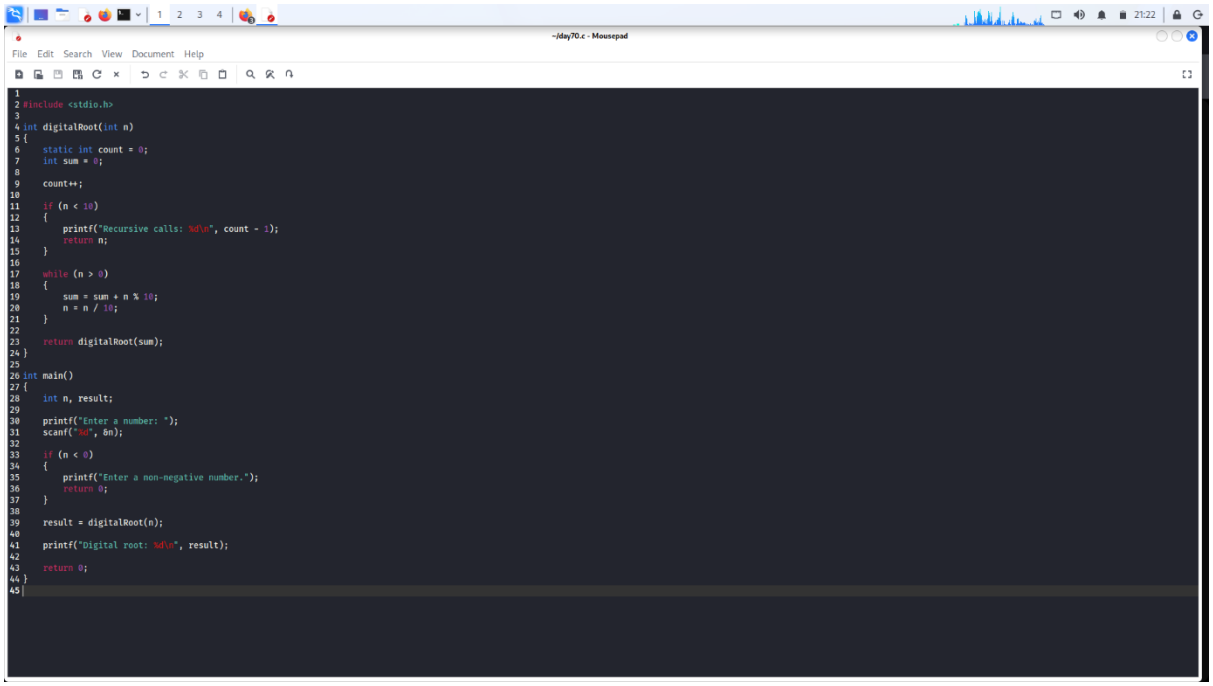
2. Problem Statement

Write a recursive function to calculate the digital root of a given number. Use a static variable to count the number of recursive calls.

3. Algorithm

1. Start the program.
2. Read an integer n from the user.
3. Define a recursive function digitalRoot().
4. Declare a static variable count to count recursive calls.
5. If the number is less than 10, return the number.
6. Otherwise, calculate the sum of its digits.
7. Call digitalRoot() recursively with the sum.
8. Display the digital root and recursive call count.
9. Stop the program.

>C PROGRAM



>OUTPUT

